

DYPAC-VLA: FULL-CONTEXT MIXED-PRECISION QUANTIZATION FOR VISION-LANGUAGE-ACTION MODELS

Anonymous authors

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ABSTRACT

Post-training quantization (PTQ) can reduce the static storage of vision-language-action (VLA) policies, yet errors that appear harmless at one layer can accumulate across action chunks and complete policy contexts. We present DYPAC-VLA—*Dynamic-Range and Prefix-Accumulated Control-aware Quantization*—a training-free mixed-precision framework that evaluates precision decisions in the complete policy. Its control metric, D_{PAC} , measures local action error, prefix accumulation, composed SE(3) pose drift, chunk-boundary discontinuity, and gripper-event timing, with a tail-risk penalty. Full-Context Precision Protection then searches native-FP16/group-64-W4 masks under an exact byte budget and accepts a change only after whole-network, cross-context adjudication. At deployment, Hessian-aware W4 weights are paired with DYRANGE-A8, which recomputes per-channel A8 scales on every forward pass; no task routing or runtime correction is used. On 50 RoboCasa365 tasks and 2,500 paired GR00T N1.5 episodes per configuration, DYPAC-VLA achieves **54.0%** success at $2.22\times$ static component compression. This is 23.6 points above QuantVLA W4A8 (Holm-adjusted $p < 10^{-4}$), while the difference from the 55.1% FP16 teacher is not significant ($p = 0.2929$). A separate $\pi_{0.5}$ study reaches $2.702\times$ compression, but is reported only as a compression anchor because its 100-episode screen does not establish a success-rate gain.

1 INTRODUCTION

Vision-language-action (VLA) models connect visual perception and language-conditioned reasoning to continuous robot control Brohan et al. (2023); Octo Model Team et al. (2024); Kim et al. (2024); Black et al. (2024); Physical Intelligence et al. (2025); NVIDIA et al. (2025). Their large multimodal backbones and iterative action generators make static storage and memory bandwidth important deployment constraints. Post-training quantization (PTQ) is attractive because it reduces numerical precision without retraining the policy Xiao et al. (2023); Lin et al. (2024a).

Quantizing a VLA policy is not simply a layerwise reconstruction problem. An individually small action error can accumulate over a 16-step action chunk; rotation errors compose nonlinearly; discontinuities appear when consecutive replans are stitched together; and a small gripper timing shift can change task outcome. Moreover, the numerical effect of quantizing one layer depends on the precision of the rest of the policy. Local activation or reconstruction scores therefore need not predict the behavior of the complete mixed-precision network.

Activation range shift creates a second failure mode. Static A8 tables calibrated on a fixed buffer can become severely mismatched to the input distribution encountered by a generative controller. In our frozen attribution, the static-A8 reference has a normalized functional objective of 467, whereas dynamic A8 differs from the FP16 control by only 0.52. This gap motivates treating range adaptation as part of the deployment method, rather than as a calibration detail.

We propose DYPAC-VLA, short for *Dynamic-Range and Prefix-Accumulated Control-aware Quantization*. The central principle is to make precision decisions in the same context in which the policy will act. First, D_{PAC} compares paired FP16 and quantized action sequences in physical coordinates. It combines local and prefix-accumulated errors with SE(3) pose, inter-replan stitch, and gripper-event terms, and adds CVaR tail risk. Second, Full-Context Precision Protection (FCP) uses counterfactual

layer flips and an exact byte-constrained candidate generator, but lets a whole-network cross-context score make the final decision. Third, the selected W4/FP16 plan is deployed using group-64 Hessian-aware weights and DYRANGE-A8, a deterministic per-forward, per-input-channel A8 rule. The final method has one global static mask and uses neither task-specific routing nor runtime correction.

The full-context search also exposes an important negative result. On GR00T, none of 116 one-layer flips improves the paired proxy around the initialized 100-W4/16-FP16 mask, and none of five additional whole-network proposals passes the minimax and component-safety rules. Thus FCP retains the initialized mask as a local optimum in the tested neighborhood; it does not claim to discover a different mask. This negative result is useful: it separates evidence for the selection rule from the final closed-loop score and prevents attributing accuracy to an unsupported mask change.

We evaluate on RoboCasa365 Nasiriany et al. (2026) using 50 paired seeds on each of 50 tasks. On GR00T N1.5, DYPAC-VLA obtains 54.0% task-macro success at 962,068,480 static component bytes ($2.22\times$ compression), compared with 30.4% for QuantVLA W4A8 and 55.1% for FP16. The improvement over QuantVLA is significant under paired McNemar testing with Holm correction ($p < 10^{-4}$); the difference from FP16 is not significant ($p = 0.2929$). On $\pi_{0.5}$, budget pruning increases W4 coverage from 80 to 121 layers and yields $2.702\times$ compression, but a 100-episode screen gives 29 versus 33 successes; we therefore make no success-rate superiority claim for that model.

Our contributions are threefold:

- (1) We introduce D_{PAC} , a control-space quantization metric that captures prefix accumulation, composed SE(3) drift, inter-replan stitching, gripper-event timing, and tail risk over complete action sequences.
- (2) We propose DYPAC-VLA, a training-free full-context precision architecture that combines exact-budget W4/FP16 search, whole-network cross-context adjudication, Hessian-aware group-64 W4, and per-forward DYRANGE-A8 without task routing or runtime correction.
- (3) We provide an artifact-audited evaluation showing 54.0% GR00T success at $2.22\times$ static compression: +23.6 points over QuantVLA W4A8 with Holm-adjusted $p < 10^{-4}$, while the difference from FP16 is not significant; we also report the $\pi_{0.5}$ compression boundary without overstating its success evidence.

2 RELATED WORK

Efficient VLA inference. Generalist VLA policies combine large multimodal encoders with continuous-control heads Brohan et al. (2023); Octo Model Team et al. (2024); Kim et al. (2024); Black et al. (2024); Physical Intelligence et al. (2025); NVIDIA et al. (2025). Efficiency methods use compact architectures, token reduction, temporal reuse, or architecture-quantizer co-design Shukor et al. (2025); Zhang et al. (2025); Yang et al. (2025); Xu et al. (2025); Fang et al. (2025). These approaches may reduce computation as well as storage, but alter the network or its execution. DYPAC-VLA instead preserves the pretrained checkpoint structure and studies static mixed-precision compression; we do not claim latency improvement for the eager fake-quant implementation.

VLA quantization. QuantVLA adapts distribution transformations to language and diffusion-action stacks Zhang et al. (2026). ActQuant allocates tensor precision from action contribution and action-aware curvature Akbari et al. (2026), while Ω -QVLA targets full-stack W4A4 with rotations and denoising-step scaling Wang et al. (2026). These methods show that control heads and time-varying distributions require specialized treatment. Our distinction is the decision context: DYPAC-VLA judges W4/FP16 changes through physical, prefix-accumulated action error in the complete policy, and deploys a single static mask without a correction selector.

Transformer and diffusion PTQ. GPTQ and AWQ use second-order information or activation salience for weight-only quantization Frantar et al. (2023); Lin et al. (2024b). Weight-activation PTQ reshapes difficult distributions through smoothing, calibration, and rotations Xiao et al. (2023); Shao et al. (2024); Ashkboos et al. (2024); Lin et al. (2024a); Liu et al. (2025); Sun et al. (2025). Diffusion PTQ further addresses timestep-dependent ranges and iterative error accumulation Shang et al. (2023); Li et al. (2023); Wu et al. (2024); Chen et al. (2025); Zhao et al. (2024). DYPAC-VLA builds

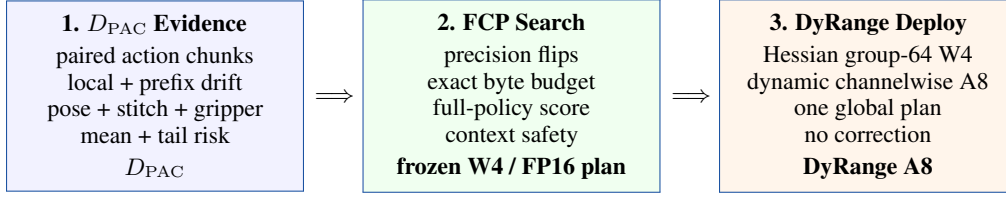


Figure 1: DYPAC-VLA architecture. Precision evidence is computed on paired physical action sequences, candidate masks are judged as complete policies under an exact byte budget, and the frozen plan is deployed with dynamic activation ranges. On GR00T, full-context adjudication retains the initialized 100-W4/16-FP16 mask rather than claiming a newly discovered allocation.

on a Hessian-aware group-64 W4 substrate. Its quantization novelty lies in full-context precision protection and dynamic per-forward activation ranges, not in a new weight-rounding primitive.

Mixed precision and behavioral sensitivity. Prior mixed-precision methods rank layers using hardware feedback, Hessian sensitivity, reconstruction loss, or diffusion-specific proxies Wang et al. (2019); Dong et al. (2019; 2020); Hubara et al. (2021); Li et al. (2021); Zhao et al. (2024). Representation similarities such as CKA and distribution divergences are useful diagnostics Kornblith et al. (2019); Yin et al. (2026), but a local feature discrepancy does not identify how errors accumulate in executed control. D_{PAC} instead operates on paired physical action sequences, and FCP reserves final authority for complete-network, multi-context evaluation under the deployment byte budget.

3 METHOD

Overview. DYPAC-VLA converts a frozen FP16 VLA into one static W4/FP16 policy. Figure 1 summarizes three stages. First, paired teacher and candidate action chunks are compared in physical control space with D_{PAC} . Second, Full-Context Precision Protection (FCP) generates byte-feasible masks from layer counterfactuals and lets complete-policy, cross-context evidence make the final decision. Third, the frozen plan is deployed with Hessian-aware group-64 W4 weights and dynamic per-forward A8 ranges. The following subsections define the optimization problem, metric, selector, and numerical deployment rule.

3.1 PROBLEM SETTING AND DESIGN CONSTRAINTS

Let $f_\theta(o, \epsilon)$ denote a frozen FP16 VLA that maps observation o and generative noise ϵ to an action chunk $A \in \mathbb{R}^{H \times d}$. The target Linear layers are indexed by $\mathcal{L}$. A mask $M \in \{4, 16\}^{|\mathcal{L}|}$ chooses signed-nibble W4 or native FP16 for every target, and $Q_M(f_\theta)$ denotes the instantiated mixed-precision policy. Our goal is

$$M^* = \arg \min_M \mathcal{R}(Q_M(f_\theta), f_\theta; \mathcal{B}) \quad \text{s.t.} \quad C(M) \leq C_{\max}, \quad (1)$$

where $\mathcal{B}$ is a frozen FP16 on-policy context buffer, $\mathcal{R}$ is a paired control-risk surrogate, and $C(M)$ is exact packed static-component storage. Unlike layerwise reconstruction, $\mathcal{R}$ is evaluated after all decisions in M have been applied to the complete policy.

The selection problem is deliberately label-free. Success and reward never enter $\mathcal{R}$; task identifiers are used only to form stratified uncertainty estimates. Selection noise is paired between teacher and candidate, while a second noise stream is reserved for post-freeze audit. The output is one mask and one activation rule shared by every task, flow step, and observation. No gradient update, extra training data, task-conditioned routing, output correction, or FP16 bypass is permitted.

These constraints separate compression design from test-time adaptation. A candidate may exploit the FP16 teacher and unlabeled on-policy observations before deployment, but it cannot observe task success or change its precision state online. This makes a selected mask auditable as a fixed model artifact and prevents a favorable task mixture from being encoded as a runtime selector.

3.2 PREFIX-ACCUMULATED CONTROL DIVERGENCE

Paired physical actions. Let $A_{r,h}^T$ and $A_{r,h}^Q$ be teacher and candidate actions at replan r and chunk index h . Both policies receive the same observation and initial action noise. We inverse-normalize the final deployed actions before comparison, giving $H = 16$ and $d = 12$ physical coordinates for both model families. For coordinate k , a single global FP16-only scale is

$$\tilde{s}_k = \max(1.4826 \text{ MAD}_k, 0.1 \text{ RMS}_k), \quad s_k = \max(\tilde{s}_k, 0.05 \text{ median}_j \tilde{s}_j, 10^{-4}). \quad (2)$$

It is estimated once over the entire selection buffer and is never re-estimated per task or sequence. This prevents a severely corrupted candidate from making its own error appear small by inflating the denominator.

Local and accumulated error. Flatten the executed actions in control-time order and define $z_t = (a_t^Q - a_t^T)/s$. We use the exactly-zero Charbonnier penalty $\rho(z) = 2(\sqrt{1+z^2} - 1)$. The local and prefix terms are

$$d_{\text{local}} = \frac{1}{Nd} \sum_{t,k} \rho(z_{t,k}), \quad d_{\text{prefix}} = \frac{1}{Nd} \sum_{t,k} \rho\left(\frac{\sum_{j=1}^t z_{j,k}}{\sqrt{t}}\right). \quad (3)$$

The $\sqrt{t}$ normalization preserves sensitivity to coherent bias: alternating errors can cancel, whereas a small error with constant sign remains visible as its prefix grows. This distinction is important for action chunks because consistent translation or rotation bias can move the end effector away from the teacher trajectory even when per-step MSE is small.

Control-specific components. Three terms capture errors that coordinatewise residuals miss. For d_{pose} , the first six action dimensions are composed as relative transforms in $\text{SE}(3)$; the teacher-to-candidate relative transform is mapped back with the Lie logarithm and normalized by $s_{1:6}\sqrt{t}$ at every prefix. The stitch term compares changes in the replan boundary jump,

$$d_{\text{stitch}} = \bar{\rho}\left(\frac{(A_{r+1,1}^Q - A_{r,H}^Q) - (A_{r+1,1}^T - A_{r,H}^T)}{s}\right), \quad (4)$$

and therefore penalizes a candidate whose individually plausible chunks do not join continuously. Finally, d_{grip} averages a soft gripper-state discrepancy and its first difference, so delayed open/close events are penalized even when continuous arm motion is accurate.

The five normalized components receive equal weight:

$$d_{\text{PAC}}^{(u)} = \frac{1}{5}(d_{\text{local}} + d_{\text{prefix}} + d_{\text{pose}} + d_{\text{stitch}} + d_{\text{grip}}), \quad (5)$$

$$D_{\text{PAC}} = \text{mean}_u d_{\text{PAC}}^{(u)} + \text{CVaR}_{0.9,u}(d_{\text{PAC}}^{(u)}).$$

Here u is a task-seed sequence of ordered replans. The outer CVaR gives the worst ten percent of sequences the same total weight as the mean, exposing rare catastrophic deviations without duplicating an observation-level tail term.

We retain a complementary functional diagnostic

$$D_{\text{func}} = d_{\text{final}} + d_{\text{kin}} + d_{\text{grip}} + 2 \text{CVaR}_{0.9}(d_{\text{solver}}), \quad (6)$$

computed on the same physical final chunks. D_{PAC} measures ordered control accumulation; D_{func} preserves a local final/kinematic view. A precision change must be conservative under both rather than choosing whichever metric is favorable.

3.3 FULL-CONTEXT PRECISION PROTECTION

Context buffer. We collect FP16 on-policy observations from 12 tasks: four each from Atomic-Seen, Composite-Seen, and Composite-Unseen. Three seeds per task and four replans per sequence yield 36 sequences and 144 observations. A hash of task and seed assigns early, middle, or late rollout windows, preventing the selector from seeing only easy initial states. Task identifiers stratify statistics, but the result remains one global mask.

Full-Context Precision Protection: freeze procedure

Input: frozen FP16 teacher, initial mask M_0 , context buffer $\mathcal{B}$, exact byte ceiling $C_{\max}$.

1. **Pair:** run teacher and each one-layer counterfactual with identical observations and noise; compute D_{PAC} , D_{func} , and component vectors.
2. **Propose:** combine conservative flip benefits with an exact sparse Pareto dynamic program; add structured single-, pair-, attention-, and MLP-family proposals.
3. **Adjudicate:** instantiate every proposal as a complete policy; retain only masks with $J < 0$ and non-positive pose, stitch, and gripper upper bounds.
4. **Audit state:** rescore the top three eligible masks on candidate-induced observations and replace J by the worse teacher-/candidate-state score.
5. **Freeze:** apply the deterministic byte/FP16/lexicographic tie-break; if no proposal is eligible, return M_0 and record the fallback reason.

Figure 2: The selector is fail-closed: the additive solver proposes masks, but only complete-policy paired evidence can change the frozen configuration.

Coordinate counterfactuals. Starting from a complete mask, round one flips every W4 target to FP16 individually; round two flips every target state around the round-one winner. Each intervention is instantiated as a real mixed-precision network and compared with its complete-mask background. The resulting conservative benefits feed an exact sparse Pareto 0–1 dynamic program under

$$C(M) = C_{\text{fixed}} + \sum_{i \in \mathcal{L}} C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}. \quad (7)$$

Layer count is not a constraint because two layers can have different tensor and metadata sizes. The dynamic program is only a candidate generator: additive counterfactual benefits cannot represent interactions between simultaneous layer changes.

Task-level minimax adjudication. For metric $q \in \{D_{\text{func}}, D_{\text{PAC}}\}$ and task cluster $g \in \{\text{all}, \text{Atomic}, \text{C-Seen}, \text{C-Unseen}\}$, let $\Delta_{q,t}(M)$ be the paired candidate-minus-background task scalar. We define

$$U_{q,g}(M) = \frac{\text{mean}_{t \in g} \Delta_{q,t}(M) + \text{SE}_{t \in g}^{\text{jack}}[\Delta_{q,t}(M)]}{\text{mean}_{t \in g} q_t(M_{\text{base}})}, \quad J(M) = \max_{q,g} U_{q,g}(M). \quad (8)$$

The leave-one-task-out jackknife makes the task, rather than an observation, the uncertainty unit. A proposal is eligible only if $J(M) < 0$ and the unnormalized one-SE upper bounds for pose, stitch, and gripper are all non-positive. Thus a favorable global mean cannot hide harm on a difficult split or safety-relevant component.

Candidate-state check and freeze. Teacher-state observations can underrepresent the states induced by a changed policy. We therefore regenerate four to eight replans for the three best eligible masks, pool candidate-induced observations, and evaluate paired teacher/candidate actions again. The final score is $\max(J_{\text{teacher}}, J_{\text{candidate}})$. Remaining ties are broken by smaller static bytes, fewer FP16 layers, and then lexicographic mask order. This makes the freeze deterministic and fail-closed.

On GR00T, all 116 valid one-layer flips have non-positive benefit relative to the initialized 100-W4/16-FP16 mask, and all five additional full-network proposals are ineligible. The unchanged context baseline is therefore frozen. This supports local optimality only in the declared counterfactual neighborhood; it is not evidence of a global optimum or a newly discovered GR00T mask. For $\pi_{0.5}$, exact-budget pruning converts the 41 lowest-benefit protected layers to W4, producing 121 W4 and 59 FP16 layers.

3.4 HESSIAN-W4 AND DYNAMIC-RANGE A8 DEPLOYMENT

Weight quantization. For each W4 layer, 256 calibration observations form FP16 input second-order statistics $X^\top X$. A GPTQ-style sequential error-feedback pass selects clipping and quantizes signed weights in groups of 64 [Frantar et al. \(2023\)](#). Two signed four-bit codes are packed per byte; per-group scales and packing metadata are retained, while the FP16 weight tensor is not. Row rotation is the identity, so the reported method does not depend on an unreported learned transform.

Dynamic activation range. Static activation tables can become mismatched as observation and denoising distributions change. DYRANGE-A8 recomputes a scale for every target Linear, forward call, and input channel:

$$s_c = \max\left(\frac{\max |x_{\dots,c}|}{127}, 10^{-6}\right), \quad q_c = \text{clip}(\text{round}(x_c/s_c), -127, 127). \quad (9)$$

The rule is deterministic and uses the input dtype. It has no calibration lookup, flow-step selector, task selector, or candidate branch; Eq. 9 is applied uniformly to every input and runtime correction is disabled.

Exact storage scope. We report the byte sum of fixed model components, packed W4 codes, per-group scales and metadata, and native FP16 tensors. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are excluded. Under this common scope the final GR00T plan contains 100 W4 and 16 FP16 targets and occupies 962,068,480 bytes. The $\pi_{0.5}$ anchor contains 121 W4 and 59 FP16 targets and occupies 1,634,828,288 bytes. These values establish static compression only; they do not imply latency or peak-memory improvements.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark and models. We evaluate GR00T N1.5 and $\pi_{0.5}$ [Physical Intelligence et al. \(2025\)](#) on the RoboCasa365 v1.0.1 target split [Nasiriany et al. \(2026\)](#): 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks. Formal configurations use 50 paired scenarios per task, official horizons, and a fresh rendered environment, producing 2,500 episodes per configuration. GR00T uses four flow steps and executes 16 actions per replan. Initial action noise is deterministically keyed by task, seed, and replan, so paired policies see the same stochastic input.

Configurations and storage. FP16 is the teacher and QuantVLA W4A8 [Zhang et al. \(2026\)](#) is the uniform low-bit reference. Uniform W6 isolates the effect of allocating more bits uniformly, while Ω -QVLA [Wang et al. \(2026\)](#) represents a more aggressive W4A4 operating point. DYPAC-VLA uses 100 W4 and 16 native-FP16 target layers for GR00T, with no output correction. Storage is exact total static model-component bytes, including fixed components and packing metadata but excluding activations, workspaces, simulator state, and fake-quant caches. The GR00T plan occupies 962,068,480 bytes, or $0.998\times$ the QuantVLA cell and $2.22\times$ compression relative to FP16.

Statistics and evidence gate. We report unweighted task-macro success, split macro success, exact paired two-sided McNemar tests, and a 10,000-draw task-then-seed hierarchical bootstrap. Reported baseline comparisons use stored Holm-adjusted values from the preregistered family. Frozen manifests bind plans, code, environments, tasks, seeds, and source hashes. A paper cell is generated only when exact coverage and artifact hashes pass the local audit; an absent or incomplete result remains “–” rather than being imputed.

4.2 MAIN RESULTS

GR00T headline. DYPAC-VLA obtains **54.0%** success, with 74.7%, 43.9%, and 40.9% on Atomic-Seen, Composite-Seen, and Composite-Unseen. QuantVLA obtains 30.4%; ours improves by 23.6 points. At the episode level there are 721 paired wins and 132 losses; the exact McNemar p -value is 6.30×10^{-99} and the Holm-adjusted value is 1.89×10^{-98} . The hierarchical 95% interval for the gain is [19.12, 28.00] points. Ours exceeds QuantVLA on 48 of 50 tasks, ties on one, and is lower on one, showing that the aggregate gain is not produced by a small set of outliers.

FP16 obtains 55.1%. Ours has 292 wins and 319 losses, with $p = 0.2929$ and a 95% interval of [−3.88, 1.76] points. The split differences are −0.9, +2.0, and −4.4 points for Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively; at task level ours is higher on 18, tied on six, and lower on 26. We therefore report a non-significant overall difference from FP16, not equivalence or superiority.

Table 1: RoboCasa365 results and exact static storage for GR00T N1.5 and $\pi_{0.5}$. Formal target-split rows use 50 paired scenarios per task.

Configuration	Low-bit layers	Atomic SR $\uparrow$	C-Seen SR $\uparrow$	C-Unseen SR $\uparrow$	All SR $\uparrow$	Size (GiB) $\downarrow$	Comp. $\uparrow$
GR00T N1.5 (<i>formal target split</i>)							
FP16	–	75.6	41.9	45.3	55.1	1.993	1.00 $\times$
QUANTVLA W4A8	116 W4	49.6	19.6	19.8	30.4	0.898	2.22 $\times$
Uniform W6	116 W6	68.4	42.9	41.8	51.7	1.109	1.80 $\times$
Ω -QVLA W4A4 [‡]	180 W4	60.1	25.9	27.5	38.7	0.599	3.33 $\times$
DyPAC-VLA (Ours)	100 W4	74.7	43.9	40.9	54.0	0.896	2.22$\times$
$\pi_{0.5}$ Physical Intelligence et al. (2025) (<i>formal baselines; ours is a quick-screen anchor</i>)							
FP16	–	58.3	14.0	2.1	26.2	4.113	1.00 $\times$
QUANTVLA W4A8	180 W4	56.1	12.8	1.8	24.8	1.388	2.96 $\times$
Uniform W6	180 W6	56.6	10.9	2.1	24.5	1.902	2.16 $\times$
Ω -QVLA W4A4 [‡]	252 W4	49.6	10.4	1.0	21.5	1.307	3.27 $\times$
DyPAC-VLA (Ours; anchor)	121 W4	–	–	–	29/100*	1.523	2.70$\times$

C-Seen/C-Unseen denote Composite-Seen/Composite-Unseen. GR00T ours uses 962,068,480 bytes; $\pi_{0.5}$ ours uses 1,634,828,288 bytes. *The 29/100 cell is a five-task quick screen, not the 2,500-episode target-split protocol, and supports no cross-row success comparison. [‡] Ω -QVLA uses model- and task-set-specific calibration. Compression denotes static component storage, not latency.

Table 2: Core mechanism ablations on the frozen 12-task, 36-sequence context buffer. These are paired offline diagnostics, not closed-loop success rates; lower objective is better.

Factor	Controlled counterfactual	Audited evidence	Decision
Range	Static A8 vs. FP16-activation control	$J = 467.52$	Reject static A8
Range	Dynamic A8 vs. FP16-activation control	$J = 0.518$	Use DYRANGE-A8
Protection	Every valid one-layer state flip	0/116 positive benefits	Keep base mask
Adjudication	5 structured mask alternatives	0/5 eligible; $J \in [3.97, 6.87]$	Reject alternatives

J is the worst normalized one-standard-error upper bound over D_{PAC} , D_{func} , and task clusters. The FP16-activation control is diagnostic only. A mask is eligible only when $J < 0$ and pose, stitch, and gripper upper bounds are non-positive.

Task complexity and storage. Relative to QuantVLA, the gains are 25.1, 24.3, and 21.1 points across the three task groups. The persistence of the gain on Composite-Unseen is relevant to the method hypothesis because held-out task compositions provide a stress test for accumulated control error. This is an empirical pattern rather than proof that any individual D_{PAC} component causes the gain. The storage comparison adds a separate constraint: ours is slightly smaller than the QuantVLA cell, whereas Uniform W6 reaches 51.7% at 1.109 GiB. Ω -QVLA is smaller at 0.599 GiB but obtains 38.7%; thus Table 1 should be read as a success–storage set of operating points rather than a single ranking.

$\pi_{0.5}$ boundary. The lower block of Table 1 adds formal $\pi_{0.5}$ baselines, each evaluated on the 2,500-episode target split. Our $\pi_{0.5}$ row has a deliberately different role: exact-budget pruning increases W4 coverage from 80/180 to 121/180 layers, reduces static bytes to 0.886 $\times$ the 80-W4 reference plan, and reaches 2.702 $\times$ compression relative to FP16. It was screened only on two ten-seed waves over five tasks. The candidate obtains 29/100 successes versus 33/100 for the 80-W4 reference, with seven paired wins and eleven losses. Because the task and seed coverage differs from the formal rows, the 29/100 cell is a compression anchor only and is not compared with their success percentages.

Table 3: Main closed-loop ablation matrix. Each contrast changes one decision while freezing the checkpoint, task–seed pairs, byte accounting, and rollout protocol. Pending cells are a preregistered experiment plan and support no current claim.

Configuration	Changed decision	Frozen controls	Status / SR
DYPAC-VLA	None	Formal 50-task protocol	Complete; 54.0
Static A8	Activation range only	Same W4 weights and mask	Planned / –
Local-MSE selection	Precision metric only	Same byte ceiling and generator	Planned / –
No full-network check	Final adjudication only	Same metric, budget, and runtime	Planned / –

4.3 MECHANISM EVIDENCE AND CORE ABLATIONS

Table 2 reports completed mechanism-level ablations on the frozen selection buffer. Static A8 is strongly refuted by the paired offline objective, whereas DYRANGE-A8 stays close to the FP16-activation control. Independently, neither a one-layer flip nor a structured full-network alternative passes the conservative selection rule. These diagnostics support range adaptation and retaining the initialized mask, but they are not closed-loop causal estimates and do not prove global optimality.

Table 3 separates the most important missing causal tests from the completed diagnostics. Static A8 tests the range mechanism while fixing the deployed weights and mask. Local-MSE selection tests whether physical prefix accumulation changes the selected plan under the same exact byte ceiling. Removing whole-network adjudication tests whether deploying the additive-program proposal directly is harmful. Each contrast reuses the same 50 tasks, 50 seeds, initial noise, horizons, and 2,500 paired episodes as the headline; task-macro success is primary and Holm correction covers the three contrasts. These rollout cells remain pending and are not used to support the present claims. Lower-priority metric, search, and budget diagnostics are specified in Appendix A.5.

Protocol separation and failure accounting. Mask selection and formal success evaluation use disjoint evidence: the former receives paired FP16 actions on an unlabeled 12-task context buffer, whereas the latter is read only after the mask, quantizers, and analysis family are frozen. Formal cells require all 50 tasks and 50 scenarios per task. A simulator crash, numerical failure, missing episode, or hash mismatch prevents a cell from entering the table; the audit never silently changes the denominator. This separation matters for a mixed-precision search because even indirect access to rollout success could turn task allocation into supervised model selection.

Deployment scope. The current implementation validates numerical behavior and exact packed storage accounting. Its eager fake-quant path retains floating-point execution, so the paper makes no end-to-end latency or live-memory claim. Fused low-bit kernels and real-robot evaluation remain future work.

5 CONCLUSION AND LIMITATIONS

We presented DYPAC-VLA, a training-free mixed-precision framework that evaluates quantization in complete policy context. D_{PAC} makes prefix accumulation, physical pose, action-chunk stitching, gripper timing, and tail risk visible to precision selection; FCP adjudicates complete masks under an exact byte budget; and DYRANGE-A8 adapts A8 scales on every forward call. The final deployment uses one static W4/FP16 mask, with no task routing or runtime correction.

On GR00T N1.5, DYPAC-VLA reaches 54.0% RoboCasa365 success at $2.22\times$ static component compression. It improves over QuantVLA W4A8 by 23.6 points, while its difference from the 55.1% FP16 teacher is not significant. The mechanism diagnostics clarify what the evidence does and does not establish: dynamic ranges avoid the severe static-A8 mismatch, and full-context search validates the initialized GR00T mask as a local optimum in the tested neighborhood.

The central result is therefore a constraint-matched operating point rather than a claim that one precision should dominate everywhere. At essentially the same static size as QuantVLA, the GR00T policy recovers most of the FP16 success rate without task routing or action correction. Uniform W6 and Ω -QVLA occupy different success–storage regions, and the $\pi_{0.5}$ anchor shows that exact-budget

mask construction transfers to a second architecture even though its success evidence does not yet support a formal claim. Keeping these roles separate is important: storage accounting establishes deployable model size, paired rollout statistics establish behavior, and offline divergence ablations explain mechanisms only within their measured context.

Our evidence is bounded in four ways. The local search does not prove global mask optimality; the core component-removal rollouts are specified but not yet complete; the $\pi_{0.5}$ result is only a 100-episode compression anchor, not a success claim; and eager fake quantization does not establish latency or live-memory gains. Completing the ablation matrix, extending the formal evaluation to additional policy families, implementing fused INT4/INT8 kernels, and validating on real robots are the next steps.

AI USE STATEMENT

Generative AI tools assisted experiment planning, software implementation and debugging, result aggregation, drafting and editing, reference and table formatting, and architecture-figure preparation. The authors reviewed all generated material, tested AI-assisted code, reran evidence and document gates, verified numerical claims against frozen manifests, and checked references against primary sources. No experimental result was synthesized by a generative model. The authors take responsibility for the final content and artifacts.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the metric, full-context search, quantization, activation, evaluation, and statistical protocols. The supplementary artifact provides content-addressed plans, manifests, task/seed coverage, table generation, and claim–evidence audits. The headline numerical cells are regenerated from frozen summaries rather than entered manually.

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A SUPPLEMENTARY EVIDENCE

A.1 FROZEN GR00T DECISION RECORD

The GR00T search starts from an initialized 100-W4/16-FP16 mask. All 116 valid one-layer flips have non-positive benefit under the cross-context score. The exact dynamic program therefore emits no beneficial half- or full-budget combination, and five additional structured proposals are evaluated as complete networks alongside the context baseline. Every alternative has a positive paired minimax objective or violates pose, stitch, or gripper safety. The frozen fallback is the unchanged context baseline. These facts support local optimality only within the declared counterfactual neighborhood.

The final plan uses signed-nibble group-64 weights, identity row rotation, per-forward dynamic A8, and no ATM, OHB, error-folding, calibration bypass, or runtime selector. Its SHA-256 is recorded in the artifact registry and checked before table generation.

A.2 PAIRED STATISTICAL DETAIL

Against QuantVLA W4A8, 1,647 of 2,500 episodes are tied, 721 favor ours, and 132 favor QuantVLA. Exact two-sided McNemar testing gives $p = 6.30 \times 10^{-99}$; Holm adjustment over the registered comparison family gives $p = 1.89 \times 10^{-98}$. The task-then-seed hierarchical bootstrap interval for the success-rate difference is $[0.1912, 0.2800]$. Against FP16, the corresponding counts are 1,889 ties, 292 wins, and 319 losses, with $p = 0.2929$ and interval $[-0.0388, 0.0176]$.

A.3 STORAGE ACCOUNTING

The GR00T FP16 static-component scope is 2,139,537,408 bytes. The final plan occupies 962,068,480 bytes, including 327,352,320 fixed bytes and packed mixed-precision target components, for $2.2239\times$ compression. This is $0.9982\times$ the QuantVLA cell of 963,772,416 bytes and satisfies the preregistered $1.10\times$ ceiling. The $\pi_{0.5}$ FP16 scope is 4,416,602,112 bytes; its 1,634,828,288-byte plan gives $2.7016\times$ compression.

These values are static packed estimates. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are outside the scope; therefore no runtime speed or peak-memory conclusion follows from the compression ratios.

A.4 $\pi_{0.5}$ CLAIM GUARD

The $\pi_{0.5}$ combined screen uses two ten-seed waves over five tasks, for 100 paired episodes per configuration. The 80-W4 reference plan succeeds in 33 episodes and the 121-W4 candidate in 29; there are seven candidate wins and eleven losses. Since this screen neither advances to the 2,500-episode matrix nor passes a superiority gate, it supports only the observed storage point. The paper registry records the generic status `quick_screen_does_not_support_success_superiority`.

A.5 EXTENDED ABLATION PROTOCOL

Table 4 contains planned lower-priority diagnostics that refine the mechanism but are not necessary to test the three central claims. Every selection ablation will be rerun from the same FP16 checkpoint and frozen context buffer. It keeps the exact-byte accounting rule, reports the resulting mask rather than forcing equal layer counts, and measures mask overlap, D_{PAC} , D_{func} , pose, stitch, and gripper upper bounds. Only variants that pass the simulator preflight proceed to the paired 50-task rollout; failures and excluded episodes are reported without imputation. Budget-curve points use independent selection runs and are not treated as repeated measurements of the headline plan.

A.6 PROTOCOL-REGISTERED LIBERO EXTENSION

Table 5 is moved to the appendix because the joint five-configuration, 4,000-episode local manifest has not passed the completion gate. Every cell is therefore marked pending, no author-reported number is imported, and the table supports no empirical claim in the current paper.

Table 4: Planned secondary ablations. These experiments are not complete and provide no numerical evidence in the current paper.

Family	Variants	Controlled readout	Status
D_{PAC} terms	Remove prefix, pose, stitch, gripper, or CVaR one at a time	Mask change, objective, component bounds	Planned
Context coverage	Omit one task split; early-only windows; no candidate-state rescore	Cross-split worst-case objective	Planned
Quantizer choices	W4 group size 32/128; non-identity row rotation; coarser A8 scale granularity	Objective, bytes, kernel metadata	Planned
Budget sensitivity	1.00, 1.05, and $1.10\times$ the QuantVLA static-byte cell	Success-storage Pareto curve	Planned

Each variant changes one factor and otherwise reuses the frozen checkpoint, task-stratified context buffer, exact-byte accounting, and paired evaluation protocol. “Planned” means specified but not yet claim-eligible.

Table 5: Appendix-only LIBERO extension under a shared local protocol (success rate, %).

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$
GR00T N1.5					
FP16	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
QUANTVLA W4A8	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Uniform W6	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Ω -QVLA W4A4	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
DYPAC-VLA (Ours)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
$\pi_{0.5}$					
FP16	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
QUANTVLA W4A8	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Uniform W6	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Ω -QVLA W4A4	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
DYPAC-VLA compression-anchor plan	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

Four official suites, ten tasks per suite, and ten held-out trials per task. All cells remain pending until the joint five-configuration manifest reaches exactly 4,000 valid local episodes; no author-reported values are imported. The official Ω -QVLA W4A4 packs are pinned. Pending cells are protocol commitments, not results.